

The Graded Cohomological Obstruction to Gluing in Set Theoretic Estimation: Literature Review, a Machine-Checked Twisted Čech Complex, and the Location of the Coupling Class

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Abstract

The project’s linearized Čech complex with *constant* coefficients (`Ste.CechComplex`) has $\check{H}^1 = 0$ identically: the full nerve of a cover is an acyclic simplex, so constant coefficients cannot see the coupling obstruction that the set-level theory (`Ste.CechObstruction`, `Ste.CechCover`) locates at the two-bit diagonal and the n -fold `allEqual`. This note attacks the open question: does a *twisted*, section-presheaf-valued Čech cohomology produce a nonzero class locating the coupling, in the sense of the Abramsky–Brandenburger–Mansfield–Barbosa cohomological-contextuality program [3, 4]? We review that program and its known *false-negative* phenomenon, then report a machine-checked answer (`Ste.TwistedCech`, Lean 4 over Mathlib, CI-verified, no **sorry**): (i) the AMB *per-section* $\check{H}^1$ -obstruction $\gamma(s)$ *degenerates structurally* in STE—the extension condition equivalent to $\gamma(s) = 0$ [4, Prop. 4.2] holds for every local section of every constraint over every cover, because STE’s section presheaf bakes global extendability into its stalks; (ii) the obstruction that *does* fire lives one degree down, in $\text{coker}(\mathcal{F}_R(V) \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}_R))$: for the diagonal with its singleton cover, the mixed family’s class in that cokernel is nonzero over *every* nontrivial commutative ring—no signed or formal linear combination of global solutions explains the mixed margins. The verdict is thus a sharp *partial positive*: a twisted cohomological invariant does locate the STE coupling, but at graded degree 0 (modulo global sections), while the degree-1 per-section obstruction of the transplanted AMB theory is provably blind here—a structural instance of the false-negative phenomenon. We close with a quantitative conjecture relating the cokernel rank to the logarithm of the forced representation blow-up.

1 The problem

Fix variables $v \in V$ with value types A_v and a constraint (feasible set) $T \subseteq \prod_v A_v$. For a cover $\mathcal{U} = \{U_j\}_{j \in J}$ of V , the *local sections* of T over a context $W \subseteq V$ are the restrictions to W of global solutions, $\mathcal{S}(W) = \{f|_W : f \in T\}$ (`Ste.VariablePresheaf.localSections`); $W \mapsto \mathcal{S}(W)$ is a presheaf under restriction. A *compatible family* assigns to each j a local section $s_j \in \mathcal{S}(U_j)$ agreeing on overlaps; it *glues* if a single $f \in T$ restricts to every s_j (`Ste.CechCover`). The set-level theory is closed in the project: rectangular (variable-separable) constraints glue over every cover (`rectangular_cechVanishesCover`); the two-bit coupling $\Delta = \{f : f(0) = f(1)\} \subseteq \text{Bool}^2$ fails to glue at the mixed family over the singleton cover (`diagonal_gluing_fails`), with the quantitative counts 4 compatible vs. 2 glued families (`diagonal_cechObstruction = 2`), scaling to $2^n - 2$ stuck families for the n -fold coupling (`cechObstruction_allEqual_ge`).

The open question addressed here: *grade* this obstruction cohomologically. `Ste.CechComplex` showed that the naive grading with constant coefficients is vacuous: for any nonempty index type

the full-nerve complex is exact at C^1 and C^2 , so $\check{H}^1 = \check{H}^2 = 0$ (`cechH1_subsingleton`)—a simplex is acyclic, and the coupling is invisible. The Abramsky–Mansfield–Barbosa (AMB) program [4] says the correct coefficients are the *linearized section presheaf*

$$\mathcal{F}_R(W) = R^{(\mathcal{S}(W))} = \text{the free } R\text{-module on } \mathcal{S}(W),$$

with restriction maps the linear extensions of section restriction. Does the resulting *twisted* $\check{H}^1$ produce a nonzero class locating the coupling? And is there a quantitative “obstruction size = smallest exact-representation blow-up” relationship? A known subtlety looms: the AMB obstruction is *sufficient but not necessary*—it can vanish on genuinely non-gluing scenarios (AMB themselves call these “false positives” of the linearized extendability test; from the obstruction’s viewpoint they are *false negatives*, and we use that convention).

2 Background: the cohomology of contextuality

2.1 Čech cohomology of a cover with presheaf coefficients

Let X be a set, $\mathcal{U} = \{C_i\}$ a cover, and $\mathcal{F}$ a presheaf of abelian groups on (the subsets generated by) $\mathcal{U}$. The Čech cochain complex of the *fixed* cover—no direct limit over refinements is taken, matching Bott and Tu [7, Ch. II] and Abramsky et al. [4, §3]—has $C^q(\mathcal{U}, \mathcal{F}) = \prod_{\sigma} \mathcal{F}(|\sigma|)$ over q -simplices σ of the nerve, with $|\sigma|$ the intersection of the members of σ , and coboundary $\delta^q(\omega)(\sigma) = \sum_k (-1)^k \rho \omega(\partial_k \sigma)$, where ρ is presheaf restriction. Then $\delta^{q+1} \delta^q = 0$ and $\check{H}^q(\mathcal{U}, \mathcal{F}) = \ker \delta^q / \text{im } \delta^{q-1}$. Two structural facts drive everything below: $\check{H}^0(\mathcal{U}, \mathcal{F})$ is exactly the group of *compatible families* of $\mathcal{F}$ [4, Prop. 3.2]; and for the *constant* presheaf on the full nerve the complex is acyclic in positive degrees (mechanized in `Ste.CechComplex`)—all cohomological information about gluing must come from the twist.

2.2 The AMB obstruction $\gamma(s)$

An empirical model in the sheaf-theoretic framework [3] is a compatible family of distributions $\{e_C\}$ over the contexts of a measurement cover; compatibility is exactly no-signalling. Its *support presheaf* $\mathcal{S}_e$ records the possibilistic content. The model is *logically contextual* at s if the support section s extends to no compatible family of $\mathcal{S}_e$; *strongly contextual* if $\mathcal{S}_e$ has no global section at all. AMB [4] linearize: $\mathcal{F}_R = F_R \mathcal{S}_e$ with F_R the free-module functor (the left adjoint to the forgetful functor $R\text{-Mod} \rightarrow \text{Set}$, 5, Def. 17), and form, for a fixed context C_1 , the *relative* presheaf $\tilde{\mathcal{F}}$ fitting in $0 \rightarrow \tilde{\mathcal{F}} \rightarrow \mathcal{F}_R \rightarrow \mathcal{F}_R|_{C_1} \rightarrow 0$. The connecting homomorphism of the long exact sequence sends (the class of) a section $s_1 \in \mathcal{S}_e(C_1)$ to

$$\gamma(s_1) \in \check{H}^1(\mathcal{U}, \tilde{\mathcal{F}}),$$

concretely: pick s_i with matching restrictions, set $c = (s_i)_i \in C^0$, and $\gamma(s_1) = [\delta^0 c]$. The pivotal characterization is [4, Prop. 4.2]:

$$\gamma(s_1) = 0 \text{ iff there is a compatible family } \{r_i \in \mathcal{F}_R(C_i)\} \text{ of the linearized presheaf with } r_1 = s_1.$$

Since a genuine (set-level) extension is in particular an $\mathcal{F}_R$ -extension with 0/1 coefficients, $\gamma(s_1) \neq 0$ implies logical contextuality at s_1 ; nonvanishing for all sections implies strong contextuality [4, Prop. 4.3]. The obstruction fires on the PR box (by hand: the forced coefficients derive $1 = 0$), and machine computations mod 2 verify nonvanishing for GHZ, the 18-vector Kochen–Specker configuration, and the Peres–Mermin square [4].

2.3 False negatives, AvN, and coefficient functoriality

Necessity fails, in two escalating stages. First, AMB’s own §5 computes the *Hardy model*: it is logically contextual at s_1 , yet $\gamma(s_1) = 0$, witnessed by an explicit $\mathbb{Z}$ -linear compatible family with a negative coefficient ($r_2 = s_6 + s_7 - s_8$)—formal linear combinations can cancel where genuine sections cannot. Their §8 then constructs a *strongly contextual* model with vanishing obstruction, and Conjecture 8.1 proposed that symmetry and connectedness restore completeness. Second, Carù [8] refuted that conjecture: strongly contextual, symmetric, connected models with vanishing obstruction exist, and the failure persists for the higher-degree generalizations $\check{H}^q$ he introduces; Carù [9] refines the invariant to cover “the vast majority” of models, with full completeness still open. On the positive side, Abramsky et al. [5, Thm. 21] delimits the reliable regime: every model admitting an *All-vs-Nothing* argument over a ring R (an inconsistent R -linear theory of its supports) is cohomologically strongly contextual, with the implication chain $\text{AvN}_R \Rightarrow \text{SC}(\text{Aff } S) \Rightarrow \text{CSC}_R \Rightarrow \text{CSC}_{\mathbb{Z}} \Rightarrow \text{SC}$; and ring homomorphisms $R' \rightarrow R$ give $\text{CSC}_R \Rightarrow \text{CSC}_{R'}$, so $\mathbb{Z}$ is the strongest coefficient choice. All known quantum strongly-contextual models are AvN, hence detected. A complementary fact from Abramsky and Brandenburger [3] calibrates what linearization can possibly see: over the *signed* reals, every no-signalling model has a global section (“negative probabilities”), so the linear invariants live strictly between no-signalling and classicality.

3 The STE transplant: two machine-checked results

The STE analogue substitutes, for the support presheaf $\mathcal{S}_e$ of an empirical model, the section presheaf $\mathcal{S}$ of a constraint T . This is a mathematically consequential substitution: *every* $s \in \mathcal{S}(W)$ extends to a global solution *by definition of* $\mathcal{S}$. In sheaf-theoretic terms the presheaf is flabby-at-sections in a way empirical supports are not; the stuck objects of STE are compatible *families*, never single sections. The module `Ste.TwistedCech` (Lean 4, Mathlib, CI-verified, no `sorry`, no axioms) builds the twisted complex honestly—coefficients $\mathcal{F}_R(W) = \mathcal{S}(W) \rightarrow_{f.s.} R$ (finitely supported functions, Mathlib’s `Finsupp`), restrictions `Finsupp.lmapDomain` along section restriction, cochain modules C^0, C^1, C^2 over an arbitrary cover, coboundaries d^0, d^1 with $d^1 \circ d^0 = 0$ (`twisted_d1_comp_d0`), and the honest quotient module $\check{H}^1 = \ker d^1 / \text{im } d^0$ (`twistedH1`)—and proves:

Theorem 1 (Structural false negative; `amb_extension_always`). *For every commutative ring R , every constraint T , every cover $\mathcal{U} = \{U_j\}$, every context index j_1 , and every local section $s_1 \in \mathcal{S}(U_{j_1})$, there is a twisted 0-cocycle $r \in \check{H}^0(\mathcal{U}, \mathcal{F}_R)$ with $r_{j_1} = \delta_{s_1}$ (the point mass). Hence the extension condition that Abramsky et al. [4, Prop. 4.2] proves equivalent to $\gamma(s_1) = 0$ is identically satisfied on STE section presheaves: the transplanted per-section $\check{H}^1$ -obstruction is uniformly zero, for reasons independent of any coupling.*

The proof is two lines of mathematics: $s_1 = f|_{U_{j_1}}$ for some $f \in T$, and the family of point masses $(\delta_{f|_{U_j}})_j$ is a twisted 0-cocycle restricting correctly. The content is the diagnosis, not the difficulty: AMB’s γ measures per-section extendability, and STE grants that for free. This is the false-negative phenomenon of §2.3 in structural form: an entire class of genuinely non-gluing scenarios (all STE couplings) on which the degree-1 per-section obstruction vanishes identically.

Theorem 2 (The coupling class fires in degree 0; `diagonal_mixed_cocycle_not_in_globalRange`, `mixedFamily_not_linearlyGlues`). *Let R be any nontrivial commutative ring. For the two-bit coupling Δ with its singleton cover, the point-mass cocycle of the mixed family ($x \mapsto \text{ff}$, $y \mapsto \text{tt}$)*

defines a nonzero class in the quotient module

$$\text{coker}(\mathcal{F}_R(V) \longrightarrow \check{H}^0(\mathcal{U}, \mathcal{F}_R)), \quad \mathcal{F}_R(V) = R^{(T)},$$

i.e., no finitely supported R -linear combination φ of the two global solutions of Δ has φ 's restriction point masses equal to the family's in both contexts. Moreover set-level gluing implies R -linear gluing (`GluesCover.linearlyGlues`) and rectangular constraints glue linearly over every cover (`rectangular_linearlyGlues`), so the class is a genuine obstruction invariant, zero on the tractable fragment and nonzero at the coupling (`diagonal_twisted_obstruction_detects`). In the mechanization the cokernel statement is rendered at representative level—"class = 0 iff the representative cocycle lies in the image submodule" (`linearlyGlues_iff_mem_range`)—because the literal quotient type over the submodule coercion $\ker d^1$ hits a typeclass-instance diamond in current Mathlib (`HasQuotient` fails to synthesize for these concrete `Finsupp/Pi` coefficient modules, while each component instance synthesizes); the submodule-level formulation is mathematically equivalent.

The proof pins coefficients exactly as in AMB's PR-box computation: evaluating the putative lift at the all-true solution forces its coefficient to be 0 (context 0) and 1 (context 1), so $1 = 0$ in R . Note the contrast with the AMB world: there, no-signalling guarantees a global section over the signed reals [3], so the analogous real-coefficient test *must* pass; here the global module is free on the solution set T itself and the margins are point masses, and even signed real coefficients fail. Theorem 2 is strictly stronger than the set-level `diagonal_gluing_fails`: "negative probabilities" cannot repair the STE coupling.

Remark 1 (Where the grading puts the class). *The compatible family's cocycle lives in $\check{H}^0(\mathcal{U}, \mathcal{F}_R)$ —Prop. 3.2 of Abramsky et al. [4] again—and the obstruction is its image in the cokernel of the global-to-local comparison map. In the long-exact-sequence picture this cokernel embeds in the $\check{H}^1$ of a relative presheaf, which is exactly how AMB's γ arises; what Theorems 1 and 2 jointly show is that after transplant to STE, the per-section relative class degenerates while the per-family comparison class survives. A direct mechanization of the relative complex and connecting map, and a proof (or refutation) that the absolute twisted $\check{H}^1$ of a pairwise-disjoint cover vanishes, are left open; the latter would make the degeneracy of Theorem 1 an instance of blanket $\check{H}^1$ -triviality for singleton covers.*

4 The quantitative question

Is there an "obstruction size = representation blow-up" relationship? The literature offers no such theorem: the established quantitative measure of contextuality is the *contextual fraction* [6]—the linear-programming distance to the noncontextual polytope, equal to the normalized maximal Bell violation—which is a convex-geometric, probabilistic-level quantity with no known link to the rank or order of the cohomological class. The nearest comparative result is Aasnaess [1], which proves the AMB Čech obstruction and the group-cohomological obstruction of Okay et al. [12] equally strong as detectors, and converts contextuality into shallow-circuit quantum advantage—computational, not representational, blow-up. The gap is real, and STE offers a clean testbed, because the project has already mechanized both sides separately: the obstruction counts (`diagonal_cekObstruction` = 2; $\geq 2^n - 2$ for `allEqual`) and the blow-up (`rectangle_superset_diagonal_encard`: every rectangle containing Δ has 4 points against $|\Delta| = 2$).

A hand computation (*not* machine-checked; stated here as data for the conjecture queue) for `allEqual` over an m -element value type with n singleton contexts, coefficients in a field: C^0 has dimension nm ; the twisted cocycle condition on the empty overlaps equates total masses across

contexts, cutting dimension to $nm - (n - 1)$; the global image is spanned by the m independent constant-solution vectors; hence

$$\text{rank coker}(\mathcal{F}_R(V) \rightarrow \check{H}^0) = nm - (n - 1) - m = (n - 1)(m - 1),$$

while the forced blow-up factor of the best rectangular over-approximation is $m^n/m = m^{n-1}$, i.e. $\log_m(\text{blow-up}) = n - 1$. For the diagonal ($n = m = 2$): cokernel rank 1, blow-up factor 2, and the two stuck mixed families map to $\pm$ the single generator.

Conjecture (outlook). *For $T = \text{allEqual } n \alpha$ with $|\alpha| = m \geq 2$ over any field, the cokernel of Theorem 2 has rank exactly $(m-1)(n-1) = (m-1) \log_m(\text{rectangular blow-up factor})$. In particular the twisted degree-0 obstruction rank is logarithmic in the representation blow-up and polynomial in the number of variables, whereas the set-level stuck-family count $(m^n - m)$ is exponential: the linear invariant is an exponentially compressed certificate of the same obstruction.*

5 Annotated bibliography

Each annotation is an original summary written for this project, with relevance to the STE transplant noted.

Abramsky and Brandenburger [3]. Recasts non-locality and contextuality as the failure of a compatible family of context-indexed distributions to extend to a global section of a presheaf on the measurement cover, proving no-signalling equivalent to compatibility and organizing Bell, Hardy, and GHZ/Kochen–Specker arguments into a strict hierarchy (probabilistic $<$ logical $<$ strong contextuality). Its coefficient-semiring analysis—global sections over $\mathbb{R}_{\geq 0}$ iff noncontextual, over the signed reals iff no-signalling—already contains the seed of the false-negative phenomenon: linearization with negatives is strictly weaker than genuine gluing. For STE it supplies the exact template: constraints as presheaves of local sections over variable covers, with the coupling as the non-gluing family (`Ste.VariablePresheaf`).

Abramsky et al. [4]. The core source for this note: defines the Čech cochain complex of the fixed measurement cover with presheaf coefficients, linearizes the support presheaf by the free-module functor F_R , and defines the obstruction $\gamma(s_1) \in \check{H}^1(\mathcal{U}, \tilde{\mathcal{F}})$ via the relative presheaf, proving (Prop. 4.2) the extension characterization on which our Theorem 1 pivots. Establishes sufficiency (nonvanishing $\Rightarrow$ contextuality), verifies the obstruction on PR/GHZ/KS/Peres–Mermin, and honestly exhibits its incompleteness on the Hardy model and on a strongly contextual model in §8, conjecturing (8.1) completeness under symmetry and connectedness. Our Lean module mechanizes its §3–§4 apparatus for STE section presheaves.

Abramsky et al. [5]. Connects the cohomological obstruction to logic: All-vs-Nothing arguments (inconsistent R -linear theories of the support) imply cohomological strong contextuality, with the chain $\text{AvN}_R \Rightarrow \text{SC}(\text{Aff } S) \Rightarrow \text{CSC}_R \Rightarrow \text{CSC}_{\mathbb{Z}} \Rightarrow \text{SC}$ (Thm. 21) and the observation that $\mathbb{Z}$ is the strongest coefficient ring; also aligns contextuality with Liar-type paradoxes and database consistency. Relevant here twice: it gives the cleanest connecting-map definition of γ , and its affine-coefficient analysis (compatibility over the empty context forces coefficient sums to one) is precisely the mass-balance phenomenon visible in our singleton-cover $\check{H}^0$ computation in §4.

Carù [8] and Carù [9]. The definitive study of the false-negative phenomenon: refutes AMB’s Conjecture 8.1 by exhibiting symmetric, connected, strongly contextual models with vanishing obstruction, and shows the failure persists for the higher-cohomology generalizations it introduces; the follow-up constructs a refined invariant complete for a large class of models while leaving a fully general complete invariant open. For STE these papers calibrate expectations: obstruction-vanishing arguments must be diagnosed structurally (as our Theorem 1 does) rather than patched degree-by-degree.

Bott and Tu [7]. Standard reference for Čech cohomology of a *fixed* cover with coefficients in a presheaf (Ch. II), including the cochain complex on the nerve and the acyclicity mechanisms that make constant coefficients on a simplex trivial—the phenomenon `Ste.CechComplex` mechanizes and that forces the twist studied here.

Abramsky et al. [6]. Defines the contextual fraction—the maximal noncontextual weight subtractable from an empirical model, computable by linear programming—and proves it equals the normalized maximal violation of generalized Bell inequalities and behaves as a monotone under the free operations of a resource theory. It is the field’s quantitative measure of contextuality, but it is convex-geometric, not cohomological; the absence of any theorem linking it (or any rank datum) to representation blow-up is the gap our §4 conjecture addresses on the STE side.

Okay et al. [12]. Develops a cohomological framework for quantum contextuality attached to the topology of Pauli/Weyl operator relations rather than to the measurement cover, subsuming Mermin-style parity proofs in both state-dependent and state-independent forms. It is the principal rival grading of the same phenomenon; its existence sharpens the question of *which* cohomology carries the STE coupling class.

Aasnaess [1]. Proves the two rival obstructions—AMB’s Čech class and the Okay–Roberts–Bartlett–Raussendorf class—detect contextuality with exactly equal strength, and generalizes the Bravyi–Gosset–König construction to manufacture shallow-circuit quantum advantage from any suitable contextual model. This is the closest extant result to a quantitative “cohomology $\Rightarrow$ separation” bridge, and it is computational rather than representational—supporting our reading that the representation-blow-up question is open.

Combettes [10]. The STE anchor: estimation as feasibility—the solution set is the intersection of context/property sets derived from data and priors, computed by projection methods. The cohomological program of this note is exactly a local-to-global analysis of Combettes-style feasibility: contexts are variable subsets, and the coupling class of Theorem 2 is a certificate that local feasibility data cannot be recombined linearly into global feasibility.

Ganter and Wille [11]. The mathematical foundations of formal concept analysis: concept lattices arise from the Galois connection between objects and attributes. The project’s constraint/feasibility polarity (`Ste.HyperFrame`) is an instance of the same adjunction, and the theories/models Galois connection used by Abramsky et al. [5, §4] to define affine closures is its linear-algebraic cousin—the natural bridge between the FCA side of STE and the cohomological side studied here.

Abramsky [2]. A survey transplanting the sheaf-theoretic and cohomological apparatus from quantum mechanics to databases, constraint satisfaction, and complexity—evidence that the AMB machinery is designed for exactly the kind of classical, constraint-based setting STE inhabits, and the methodological license for this project’s transplant.

6 Verdict and outlook

Verdict: partially solved, with a machine-checked negative component. A twisted, section-presheaf-valued Čech invariant *does* locate the STE coupling obstruction—but not where the transplanted AMB theory grades it. The per-section $\check{H}^1$ -obstruction γ degenerates identically on STE section presheaves (Theorem 1, CI-verified): a structural false negative explained by the flabbiness of `localSections`. The surviving invariant is the family-level class in $\text{coker}(\mathcal{F}_R(V) \rightarrow \check{H}^0)$, nonzero at the diagonal over every nontrivial ring (Theorem 2, CI-verified), vanishing on rectangles over every cover. Open: mechanizing the relative complex and connecting map; the vanishing of the absolute twisted $\check{H}^1$ for pairwise-disjoint covers; nonzero twisted $\check{H}^1$ classes for covers with substantive overlaps (the XOR/parity triangle is the natural candidate, being AvN over $\mathbb{Z}_2$); and the rank conjecture of §4.

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